

isobaric specific heat capacity $[J \cdot g \cdot K^{-1}]$
 mass $[g]$

$$1. q_M = c_{p,M} \times m_M \times \Delta T$$

temperature difference $[K]$

$$2. \Delta T = T_f - T_i$$

3. Assumption: the heat of the metal was solely absorbed by the water.
 I.e: heat absorbed by all other components are negligible.

$$\Rightarrow q_M = -q_W$$

4. from ③ it follows that

$$c_{p,M} \times m_M \times \Delta T_M = -c_{p,W} \times m_W \times \Delta T_W$$

$$5. c_{p,M} = \frac{-c_{p,W} \times m_W \times \Delta T_W}{m_M \times \Delta T_M}$$

6. from Dulong & Petit relationship

$$M_M [g \cdot mol^{-1}] \approx \frac{25 [J \cdot g \cdot K^{-1} \cdot mol^{-1}]}{c_{p,M} [J \cdot K^{-1}]}$$

$$\begin{cases} 25 \text{ comes from} \\ M_{c_{p,M}} = 3k_B N_A \\ k_B N_A = R \end{cases}$$

$$7. \text{By } \textcircled{6} \rightarrow \textcircled{5} \quad M_M \approx \frac{25 \times m_M \times \Delta T_M}{-c_{p,M} \times m_W \times \Delta T_W}$$